

Cloud computing AWS (Amazon Web Services)

Cloud computing is the delivery of computing resources such as **servers, storage, databases, networking, and software over the Internet** instead of using physical systems locally.

In simple words, cloud computing allows us to **use IT resources whenever we need them without purchasing and maintaining the physical hardware ourselves.**

Example

Instead of buying a physical server for a company, we can create a virtual server on the cloud and use it through the Internet.

Real-life example:

Google Drive provides cloud storage. We can store files online and access them from different devices.

Advantages of Cloud Computing

- Cost-effective
- Scalable
- Flexible
- Available from anywhere
- No need to maintain physical hardware
- Easy backup and recovery

AWS (Amazon Web Services) is a cloud computing platform provided by **Amazon**.

AWS provides various cloud services such as:

- Compute
- Storage
- Database
- Networking
- Security
- Monitoring
- Identity and Access Management

AWS allows organizations to create and manage IT infrastructure through the Internet.

An **AWS account** is used to access and manage AWS cloud services.

After creating an AWS account, we can use the **AWS Management Console** to create and configure services such as EC2, S3, RDS, VPC, and IAM.

AWS Management Console

The **AWS Management Console** is a web-based interface used to create, configure, monitor, and manage AWS resources.

Region

A **Region** is a geographical location where AWS has multiple data centers.

Example:

- Mumbai – ap-south-1
- Singapore – ap-southeast-1
- US East – us-east-1

Availability Zone

An **Availability Zone (AZ)** is an isolated data-center location within an AWS Region.

For example:

Mumbai Region → Multiple Availability Zones

Examples of Cloud Providers:

- AWS – Amazon Web Services
- Microsoft Azure
- Google Cloud Platform (GCP)

2. Cloud Computing Service Models

Cloud services are mainly divided into **three service models**:

1. **IaaS – Infrastructure as a Service**
2. **PaaS – Platform as a Service**
3. **SaaS – Software as a Service**

2.1 IaaS – Infrastructure as a Service

IaaS provides basic computing infrastructure such as:

- Virtual machines
- Storage
- Networking
- Operating systems

The customer manages the **OS, applications, and data**, while the cloud provider manages the physical infrastructure.

Example: AWS EC2

Simple example:

It is like **renting a computer/server** from the cloud instead of buying one.

2.2 PaaS – Platform as a Service

PaaS provides a ready-made platform for developers to **develop, test, and deploy applications**.

The cloud provider manages the infrastructure and operating system, while the developer mainly focuses on the application and data.

Examples:

- AWS Elastic Beanstalk
- Google App Engine
- Microsoft Azure App Service

Simple example:

It is like **renting a ready-to-use development environment**.

2.3 SaaS – Software as a Service

SaaS provides **ready-to-use software applications over the Internet**.

Users don't need to install or maintain the underlying infrastructure.

Examples:

- Gmail
- Google Drive
- Microsoft 365
- Salesforce

Simple example:

You simply **use the software** without managing the server or operating system.

Cloud deployment models: describe **where and how cloud infrastructure is deployed and who can access it**.

There are four commonly discussed models:

1. **Public Cloud**
2. **Private Cloud**
3. **Hybrid Cloud**

4.1 Public Cloud

A **public cloud** is cloud infrastructure provided by a third-party cloud provider and shared among multiple customers.

Examples:

- AWS
- Microsoft Azure
- Google Cloud

Example:

A company creates an EC2 instance on AWS instead of maintaining its own physical server.

Advantages

- Lower infrastructure cost
- Easy scalability
- No need to maintain physical hardware
- Pay-as-you-use options

4.2 Private Cloud

A **private cloud** is dedicated to a **single organization**.

The infrastructure may be hosted within the organization's own data center or by a third party.

Example:

A large organization creates a dedicated cloud environment for its internal applications and data.

Advantages

- Greater control
- Customization
- Suitable for specific security and compliance requirements

4.3 Hybrid Cloud

A **hybrid cloud** combines **private cloud/on-premises infrastructure with public cloud services**.

Example

A hospital may keep sensitive patient information in its private infrastructure while using AWS for other applications or additional computing capacity.

Private Infrastructure + Public Cloud = Hybrid Cloud

Advantages

- Flexibility
- Scalability
- Can keep certain workloads in private infrastructure
- Can use public cloud resources when required

1. AWS Account Creation

Steps

1. Open the official **AWS Sign-Up page** and click **Create an AWS Account**.
2. Enter your **email address** and **AWS account name**.
3. Verify your email using the **OTP/verification code**.
4. Create a **strong password**.
5. Enter the required **personal/contact details**.
6. Verify your **mobile number**.
7. Add a valid **payment method** supported by AWS.
8. Select the appropriate **AWS plan/support option**. For learning, choose the basic/free option when available.
9. Complete the registration process.
10. Wait for the account to be activated.
11. Sign in to the **AWS Management Console**.
12. Open **Billing and Cost Management** and review the available settings before creating resources.

2. Enable MFA for Root User

1. Sign in to the **AWS Management Console** as the **root user**.
2. Click your **account name** at the top-right corner.
3. Select **Security Credentials**.
4. Find **Multi-Factor Authentication (MFA)**.
5. Click **Assign MFA device**.
6. Enter an **MFA device name**.
7. Select **Authenticator app**.
8. Install an authenticator app such as **Google Authenticator** or **Microsoft Authenticator** on your phone.
9. Scan the **QR code displayed by AWS**.
10. The authenticator app generates a **six-digit OTP**.
11. Enter the OTP in AWS.
12. If AWS asks for two consecutive codes, wait for the next code and enter it as instructed.
13. Click **Add MFA** to complete the setup.

MFA: Password + Authenticator Code = Additional Account Security.

aws

Try AWS at no cost for up to 6 months

Start with USD \$100 in AWS credits, plus earn up to USD \$100 by completing various activities.

Sign up for AWS

Root user email address
(used for account recovery and as described in the AWS Privacy Notice)

AWS account name
(Choose a name for your account. You can change this name in your account settings after you sign up.)

Verify email address

OR

Sign in to an existing AWS account

This site uses essential cookies. See our cookie notice for more information.

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Free Tier offers

All AWS accounts can explore 3 different types of free offers, depending on the product used.

- Always free**
Never expires
- 12 months free**
Start from initial sign-up date
- Trials**
Start from service activation date

Sign up for AWS

Contact Information

How do you plan to use AWS?

- ☐ Business - for your work, school, or organization
- ☐ Personal - for your own projects

Who should we contact about this account?

Full Name

Phone Number

Country or Region

Address

City

State, Province, or Region

Postal Code

☐ I have read and agree to the terms of the AWS Customer Agreement

Continue (step 2 of 5)

Free (6 months)

Learn, experiment, and build prototypes

- ✓ Receive up to \$200 in credits
- ✓ Includes free usage of select services
- ✗ Workloads scale beyond credit threshold
- ✗ Access to all AWS services and features

ⓘ After the 6-month free period or when all credits are used, you can choose to upgrade to a paid plan. Otherwise, your account closes automatically.

Choose free plan

Paid

(DevOps) production-ready workloads

- ✓ Receive up to \$200 in credits
- ✓ Includes free usage of select services
- ✓ Workloads scale beyond credit threshold
- ✓ Access to all AWS services and features

ⓘ After all of your credits are used, you are charged using pay-as-you-go pricing.

Choose paid plan

[View additional details](#)

UPI AutoPay information

Use your preferred UPI app to setup automatic payments. You can cancel AutoPay at any time. [Learn more](#).

Automatic payment limit

₹15,000

You will need to scan the QR code which will appear on the next page to setup UPI AutoPay.

Billing address

☒ Use contact address
haridwar
haridwar, UR, 249402
IN

☐ Use a new address

ⓘ A ₹2 refundable transaction is required to verify your payment method.

[Continue from Step 5 of 5](#)

Primary purpose of account registration

Choose one that best applies to you. If your account is tied to a business, select the one that applies to your business.

Academic

Ownership type

Choose your ownership relation to the account. Based on your selection, you may be asked to complete additional customer verification steps.

Individual

India document type [Info](#)

To verify your identity, the name on the document must match the name that you chose.

Aadhaar

ⓘ Click below to complete Aadhaar verification through DigLocker. On the DigLocker portal, select Aadhaar as your document type.

Get DigLocker URL